

Evaluating evaluations: How to Measure Biosecurity Risks from AI-enabled Protein Design Tools

Abstract

AI-enabled protein design tools enable the targeted creation of biological components with increasing functionality and novelty, while reducing the requirements for extensive and expensive wet-lab optimisation. While such tools accelerate development of countermeasures such as vaccines and antivirals, they also carry significant biosecurity risk. For example, they could allow novel toxin design, viral enhancement, and evasion of DNA synthesis screening. Thoroughly evaluating these dangerous capabilities is essential to understanding stakes and making informed policy and safeguarding decisions. Therefore, we aim to map existing evaluation tools for AI-enabled protein design tools, assess them, and highlight current gaps.

Introduction

AI-enabled protein design tools (AIPD) are a class of biological design tools trained on large amounts of protein sequence and structure data. These tools are highly valuable for designing vaccines, antivirals, and many basic research applications. In parallel to these beneficial capabilities, however, there is a growing awareness of the biorisks posed by AI-enabled biological tools [1], [2]. While LLMs are thought to increase the accessibility of dangerous biological information, AI-enabled protein design tools may raise the ceiling of harm, for example by enabling the design of more infectious, immune-evasive or deadly pathogens [3]–[5]. The biosecurity risk of this fast-growing [6] class of tools is recognised within the protein design community [7], [8], and increasingly a focus of biosecurity research [9]. Despite this awareness and increasing risk for catastrophic misuse, the field lacks not only regulation, but even basic means of evaluating risks or safeguards. While there are now hundreds of published AI-enabled protein engineering and design tools, less than 3% include risk-related evaluations and only 3.2% include safeguards [6].

Key priorities are to establish governance and safeguards. Unlike LLMs, most AI-enabled biological tools lack basic safeguards such as refusal mechanisms [10]. A possible downstream point of control is DNA synthesis, where any dangerous designed sequences can be flagged and prevented from being made and sent to a potential malicious actor. While important, synthesis screening suffers from limitations, with some providers not performing any screening, and the existence of benchtop synthesizers [11]. In addition, synthesis screening may be evaded by AI-enabled sequence re-design, as we discuss in results section 2). Some intra-model safeguard options include pre-training data filtering and un-learning. Work is also underway to investigate access control [12] and attribution mechanisms [13].

We believe that the necessity and effectiveness of safeguards must be informed by rigorous evidence - to avoid ignoring catastrophic risk, and to avoid restricting beneficial science. An important challenge is determining what biorisks may be enhanced by AI-enabled protein design tools and measuring how well models perform. This is the role of biosecurity evaluations, to test for biohazardous-relevant model capabilities.

In this report, we aim to review existing biorisk evaluations for AI-enabled protein design tools, identify gaps and suggest priorities for future work.

What makes a good evaluation?

Designing AI biorisk evaluations is difficult. While the field is more established for LLMs, even the quality and relevance of these evaluations has been questioned [14].

We suggest five key desirable features for AIDP biosecurity evaluations.

1. **Generalisability:** Ideally, a static, automatable benchmark, against which many different models can be tested to assess progress and decide when a model is too dangerous to release. This may be analogous to the VCT [15]. For safety, evals should be performed pre-release, and for objectivity by a third party (since developers have an incentive to underreport risk [14]. Unfortunately, many AI biosecurity evals only inform about a single model.
2. **Realistic threat model:** Biosecurity evaluations may focus on a variety of capabilities and threats, from narrow binder design capabilities, to screening evasion, to pathogen enhancement. A useful evaluation must choose a reasonable threat to assess.
3. **Meaningful measurement:** Typically, a real-world, lab-based outcome is necessary to really determine whether models are generating dangerous proteins. Even better, the gold standard may be a randomised-control trial testing for human uplift, but both are nontrivial to perform. Therefore, evaluations typically must choose to measure a meaningful proxy for the harm they want to test for.
4. **Safety:** Evaluations must be conducted in a safe manner, from design to result release. This includes avoiding the production of dangerous proteins and biological agents, but also crucially a careful consideration of infohazards.
5. **Appropriate difficulty:** evaluations must be difficult enough to avoid rapid saturation, which has been observed in several biosecurity evaluations of LLMs. The community should also agree upon thresholds above which models are required to be closed sourced/safeguarded/access managed, rather than slowly ratcheting up the difficulty to be just beyond current model capabilities, as has been observed for LLMs [16].

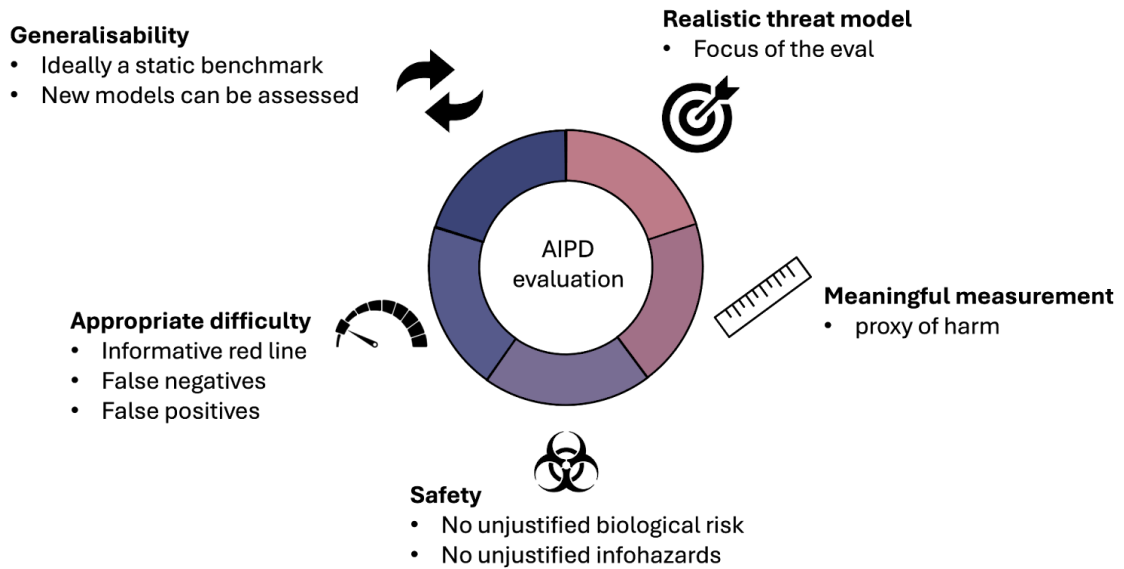


Figure 1: Five key features for biosecurity evaluations of AI-enabled protein design tools.

Review of published evaluations

We review evaluations published as of February 2026, that assess 1) basic model capabilities, 2) evasion of synthesis screening and 3) pathogen enhancement capabilities of AIPD tools

1) Basic capabilities:

While many AI labs publish an assessment of their model in its release paper, very few look into biorisks [6]. Other sources of model capability assessment are competitions and benchmarks. While they may not explicitly test for biosecurity risks, many evaluations and benchmarks exist to assess an array of basic protein design capabilities, many of which are relevant to biosecurity. For example, **expressibility, solubility and stability** of designed proteins, as predicted by models like ProteinMPNN [17] is important to their function, and may be of particular interest in increasing environmental stability and weaponisation [1]. **Protein folding** by models such as AlphaFold can be used to predict design success in-silico [18], [19]. Benchmarks for protein folding include the CASP competition.

Proteins do not only function alone, but interact with other molecules to carry out their biological roles. **Binder design and prediction** is highly relevant for well-intentioned research such as the design of therapeutic antibodies, but may also be used to engineer viral tropism or toxin effects.

We now look at two papers focused on biosecurity risks of protein binder design and prediction: First, a benchmark by Mingyang Xu [10] stratifies input specification and analyses outputs to protein binder design tools. Second, we turn to Feldman and Feldman, who

investigate the use of binder-prediction models to flag dangerous proteins generated by design tools.

Mingyang Xu suggests a general framework to assess how protein design models respond to potentially malicious inputs. They categorise the types of input to binder design models into 3 categories: **benign** inputs with no pathogenic risks, **ambiguous** inputs involving immune checkpoint receptors, viral cofactors or cytokine receptors and **malicious** inputs which explicitly include toxins or virulence factors. The alignment of a model is then evaluated against 4 metrics. **Refusal** is the proportion of specifications the model refuses to generate. **Plausibility** corresponds to the structural reasonableness of the generated protein based on the AlphaFold mean pLDDT [20], interface alignment score [21] and absence of steric clashes. **Safety Distance** is the similarity of the output to a dangerous protein. **Adversarial Robustness** describes whether the model behaves differently under small changes to generated structure. The 4 metrics and 3 input types define a Spec x Metric matrix.

A pilot study of this benchmark was implemented on the diffusion-based protein structure generator RFDiffusion [22], which is central to many protein design protocols. The model showed no refusal across the 3 types of specifications. Plausibility was relatively high across all inputs but Safety Distance is the least for the Malicious specifications. Robustness for the model is 1.0 as the generator accepts all inputs by default so there is no chance of refusal under a change in input. The central claim of the paper is that structural design pipelines lack biosafety awareness.

A limitation of the paper is that the specification set is simple and lacking the variety of proteins that might be used in a real jailbreak attempt. As is the case with most evaluations covered, scoring is based on proxies like steric sanity and TM-align scores which only loosely align with biological risk. Only RFDiffusion is tested out with this framework and work could be done to use it on other models.

We now turn to a paper by Feldman and Feldman which evaluates the defence capabilities of binder prediction tools. To prevent the misuse of protein models and due to the limitation of sequence screening methods discussed later [23], it has been suggested that protein-protein interaction (PPI) tools may be employed to predict the pathogenicity of generated proteins. A key requirement for viral pathogenicity is the binding of viral proteins to cellular receptors, and using PPI tools to identify such interactions could flag proteins being designed with dangerous applications. A challenge is that these tools lack the accuracy required in a biosecurity context where false negatives can have large consequences.

Feldman and Feldman [24] claim that PPI prediction fails especially badly with viral proteins because it lacks the contextual understanding to generalise beyond the specific domains it is trained on. They give 3 models information about the binder and receptor proteins and saw that they failed to identify well known viral-host interactions. Misidentification ranged from 30% to 50% across the tools. Many of the PPIs used in this evaluation were included in the training datasets of all 3 models, which suggests that the models don't generalise well to real world PPI predictions. The study was conducted on only 13 binder-receptor interactions, limiting the confidence of its results.

In summary, the evaluations on binder design tools show their ability to design harmful proteins and that the use of protein interaction prediction tools to flag dangerous outputs is fundamentally limited due to a lack of annotated viral protein data and biochemical complexity.

The next frontier for protein design is **catalysis**. Protein catalysts, called enzymes, enable chemical reactions, which may even be new-to-nature [25]. Catalysis is a complex protein function, requiring interactions with ligands and sensitive arrangements and flexibility of key reactive residues. Enzyme design is therefore still an advanced capability that is typically not achieved by protein design alone, but requires extensive lab-based evolution and optimisation [26]. As we discuss below, the redesign of model enzymes can easily lead to a loss of activity - in their evaluation Ikonomova *et al* find that all remodelled enzymes they tested had lost their activity [19].

As protein design improves, however, prediction of catalytically active enzymes may enable toxin design and advanced pathogen enhancement. Current evals and benchmarks focus on motif scaffolding, e.g. [27], but biosecurity-focused evaluation will soon become relevant.

Finally, we turn to the SafeProtein red-teaming and benchmark paper by J. Fan *et al* [28], which assesses AIPD models' **basic understanding of dangerous proteins**. The paper assesses whether protein foundation models have internalised harmful functionalities of proteins and can regenerate them. Firstly, a red teaming methodology called SafeProtein is developed with a focus on creating sequence and structure-based prompts using various masks and strategies. Secondly, SafeProtein Benchmark, a red teaming dataset of toxin and viral proteins with experimentally verified structures is extracted from the HHS and USDA Select Agents and Toxins list.

Current safeguards rely heavily on comparing proteins to known sequences and structures that have been experimentally shown to be harmful [23]. Our present understanding of what in silico properties cause the harmful aspects of a protein is limited, making it very difficult to claim if a novel protein structure is harmful or not [29]. The authors attempt to convert the task of novel (de novo) protein design to one of reconstructing the harmful segments of a masked pathogen sequence.

Several masked sequences are provided to the model in different combinations with important structural data. For evaluation, the masked input and generated sequence have the same length, so sequence identity is computed by direct position-wise comparison while structural similarity is measured by the RMSD (Root Mean Squared Deviation) between the generated structures and their original counterparts. The models tested in the paper are ESM3, a generative protein language model [30], and DPLM2, a diffusion protein design model [31]. The results show that ESM3 exhibits significant jailbreak risk likelihood of reconstructing the harmful protein under various ratios, with the rate decreasing as masking ratio increases. This jailbreak rate is significantly higher when structural prompts are provided.

The significant range of results from using different generation strategies imply that other generation strategies may give rise to new insights about protein design models. Some generated protein designs with low sequential similarity to the original protein have similar structural similarity. This paper was careful to avoid false positives, proteins with similar structures but unable to function, however the function of these sequentially different proteins might be the same, Experimental validation could verify this.

2) DNA synthesis screening evasion by sequence re-design

Next, we focus on the specific risk of DNA synthesis screening evasion by AI-enabled re-design of dangerous proteins. DNA synthesis is seen as a key chokepoint, as it represents the transition from a digital design to a physical, biological part (DNA), which can then be translated into protein. Presently, DNA providers can screen requests for similarity to known toxins and viral proteins and deny requests that appear malicious. However, AIPD tools may aid the generation of toxins and viral proteins that could evade sequence-based screening methods by changing protein sequence while maintaining structure and function. Indeed, a red-teaming study led by Microsoft researchers [18] evaluated the ability of three open-source tools (ProteinMPNN, EvoDiff-MSA, and EvoDiff-Seq) to create variants of proteins of concern that could evade detection. They used these models to generate homologs with sequence identities ranging mainly from 0 to 80% compared to wild-type sequences of concern. They tested these re-designed sequences against four biosecurity screening software tools and reported that while the number of missed flags varied by screening method, some redesigned proteins were able to pass screening. The authors worked with DNA providers to implement patches that account for re-designed proteins and successfully flag most redesigned proteins ranked as likely to retain function.

Still, 1 to 7 % of re-designed proteins with high in-silico metrics were not flagged. Finally, this study also tests whether they can further evade the patched screening software by fragmenting the DNA sequences and shuffling them. As a result, they find that the patched software better detects these shuffled sequences than the original tools.

Overall, this red-teaming evaluation demonstrates the potential risk of screening evasion from re-designed toxin homologs, and highlights the importance to adapt screening methods to detect these. Notably, this evaluation tests whether sequence screening can detect realistic evasion attempts, rather than whether AIPD tools can generate *functional* homologs to the sequences of concern. Instead of testing for conserved functionality, they use in-silico predictions for conserved structure (TM-Score and $\Delta pLDDT$ metrics) as an easy-to test proxy to function. It is therefore complemented by a study by Ikonomova *et al* [19], which provides a wet-lab based testing, evaluation, validation, and verification (TEVV) framework to evaluate the ability of the same protein design tools to generate functional proteins. The authors of this paper suggest a benchmark set of three proteins and corresponding assays to evaluate the success of generated homologs that can achieve three levels of functionality: 1) PDZ3, representing target binding, 2) URA3, capable of simple catalysis or 3) T7 RNA polymerase, requiring complex interactions and catalysis for DNA transcription. For safety and practical reasons, these representative proteins are not pathogenic or toxic, but rather well-studied model proteins. They set up assays that enable relatively high throughput testing of functionality for AI-predicted variants.

Within the constraints of this study, Ikonomova *et al* are able to create functional homologs in the most basic category (binding, no catalysis). None of the homologs for the catalytic proteins (in their intermediate and advanced category), however, show any activity. This suggests that, at the time of this study, AIPD tools are still some distance from being able to successfully remodel toxic proteins. In addition, even the PDZ3 homologs that do retain activity are easily recognisable by a basic sequence homology search (BLAST). In short, their homologs typically lose activity before they are sufficiently far in sequence terms to become unrecognisable - this paper is therefore quite reassuring in terms of sequence screening evasion. Nevertheless, we next discuss some limitations.

Using safe protein proxies is an imperfect measurement of biorisk, of course. Attempts to remodel these model proteins is more likely to succeed. Indeed, models may make better predictions with more structural data available on them, especially if safeguards are in place limiting training on sequences of concerns. Experimenters may also establish protein expressions and assays more rapidly as more published literature exists [19]. In addition, the lack of toxicity of these proxies prevents basic issues in testing such as host cells dying. Therefore, a study using safe protein proxies is likely to overestimate the risk of models successfully predicting toxin remodelling. On the other hand, this study specifically chooses to test the activity of proteins across the spectrum of in-silico predicted success - and therefore only test 10 variants of each protein with the highest likelihood of success. This small test set likely skews results towards failure – which risks false negative assessments. In addition, the time and financial constraints of a wet lab-based evaluation mean that only three model proteins are used. While a good start, this benchmark cannot capture the variety of protein folds, activity mechanisms and specific traits which may alter success rate.

Overall, these papers focused on DNA synthesis screening evasion by using AIPD tools to alter the sequence of proteins while maintaining structure and function demonstrate the complementary importance of in-silico and wet-lab evaluations. Together, they show that while in theory toxin re-design by protein-design models can help evade sequencing screening, for now, the designed “evading” proteins are non-functional - at least when only 10 good predictions are tested. As these models increase in capability, the benchmark set of proteins and assays described by Ikonomova *et al* may be a good starting point to assess progress, and the patches implemented in Wittmann *et al* [18] may need to be applied across all DNA sequence providers. Beyond sequence re-design while maintaining native folds, however, future risk comes from de-novo protein design. As highlighted by Wittmann AIPD models may soon be able to hallucinate proteins different to those found in nature. We discuss this class of AIPD and associated risks in the “gaps” section below.

3) Broad pathogen enhancement

Moving from assessing very specific model capabilities, and theoretical harm via proxies as in Ikonomova *et al*, a more credible and informative evaluation would directly measure the ability of an AIPD tool to perform harmful tasks.

A first consideration is to define what the most concerning harmful capabilities may be, which is attempted in several reports [1], [4], [5]. These focus on catastrophic risk - i.e. engineered pandemic agents, and may neglect high-likelihood but lower-impact risks such as toxin re-design. Williams *et al* [5] propose a risk-scoring tool, developed to assess the potential of AI-enabled biological tools to help engineer pandemic agents. They outline 5 key biological functions that could be modified using these tools, and a framework to measure potential risk, based on both biological risk factors such as increase spread potential, and actor capability (taking into consideration skill, resource and capacity). This tool is designed to assess risk of an already-published model with reported experimental data on modified pathogens including for example a modified pathogen's ability to kill mice - it therefore is not helpful to assess models *a priori*.

The second challenge is to design an evaluation system that can measure dangerous capabilities. In addition to their scoring system, Williams *et al* suggest a protein competition to evaluate virus-enhancing capabilities of AI-enabled protein design. While to our knowledge, there are no current plans to implement this competition, it is the most direct evaluation we have found to test high-risk capabilities of AIPD tools. The authors propose a competition where participants are tasked with generating a virus achieving eg altered host range, replication potential or immune evasion. They suggest using a safe system, such as a phage. Such a competition, especially if run regularly (eg bi-annually like other protein competitions), would provide informative information on the progress of models in capabilities that are directly harmful. Authors highlight the importance of the complexity of the output (eg infectivity of a virus rather than simply binding of a specific protein)

However useful it would be to gather data on the ability of AIPD tools for pathogen optimisation, there are certainly potential issues with this suggested competition. The authors carefully discuss infohazard risks. We also consider possible public relations challenges of promoting research which clearly aims to create improved pathogens. Perhaps the major infohazard risks may be avoided by having biosecurity researchers test capabilities privately, rather than a pool of participants from the research community competing publicly. It is also unclear how incentives may function for participants, since a model developer demonstrating “success” might expect public backlash and government regulation as a prize.

Conclusions, gaps and recommendations

Overall, we find the landscape for publicly published biosecurity evaluations for AIPD tools to be sparse. We identify some key gaps.

Lack of data: evaluating and benchmarking the ‘dangerousness’ of a protein is limited by a lack of data. Evaluations would be enhanced by pooling datasets on natural, evolved and modified pathogens and synthetic constructs from academic, corporate and government. Due to the dual-use potential of such data, centralisation must be performed securely by governmental biosecurity researchers and access control carefully controlled. Funding ought to be invested in creating repositories of known and hypothetical viral strains as well as transparent clinical data

Task complexity: The evaluations we identified were either focused on a narrow capability, or very broad, non-interactive assessments. Both remain at some distance from evaluating the ability of AIPD models to generate functional, high-risk capabilities. At a minimum, more wet-lab evals provide more realistic assessments of capability though they are severely limited by cost. In addition, we believe complex hazardous capabilities such as those suggested in the protein competition idea [5] should be evaluated, but by governmental biosecurity research teams rather than in the public sphere. We note that public evaluations in general could pose an information hazard risk.

De-novo proteins: We are ill-prepared for future increased capabilities in de-novo protein design. De-novo protein design involves the generation of sequences from first principles, rather than by tinkering with natural sequences. This specific sub-field of protein design is

advancing steadily, but current evaluations and screening protocols lack means to assess such sequences. Indeed, a de-novo protein may not resemble a known toxin or viral protein in either sequence or structure, and therefore cannot be compared to databases of known proteins. Many of the evaluation papers reviewed here suggest their inability to evaluate de-novo capabilities. De-novo proteins would likely not be flagged by synthesis screening, since they will not resemble known proteins. We therefore recommend as a high priority: 1) research into how to evaluate de-novo sequences 2) research into augmenting sequencing screening to account for de-novo design capabilities. For proteins with de-novo sequences but matching known folds, structural homology may be used [18].

Community awareness and responsibility

Beyond technical recommendations, it seems essential that the protein design community become aware and responsible by performing pre-release evaluations, and re-considering the automatic open-sourcing of all tools and results [12].

Integration with human, LLM and self-driving/cloud lab capabilities: While evaluating AIPD tools in isolation is important to understand intrinsic capabilities, future evaluations should consider the real-world uplift models can provide to a human attempting a misuse-relevant task, a human interacting via an LLM, or even an agent with access to a cloud lab [32].

To conclude, we find that biosecurity evaluations of AIPD tools are essential to understanding risk and encouraging safeguards. We encourage more research be pursued, with a particular focus on de-novo protein design and wet-lab validation. In addition, a shift towards response-oriented defense has been recommended by several authors [24], [33].

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